

JKBOSE 2026

CLASS 11TH

Bhai'charya
ON
TOP.

CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES

LECTURE-1

CHEMISTRY

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Topics To Covered

1 STABILITY OF HALF FILLED AND FULLY FILLED ORBITALS

STR. OF ATOM

2 SIGNIFICANCE OF CLASSIFICATION OF ELEMENTS

Classification of
Elements &

3 BRIEF HISTORY OF PERIODIC TABLE

Periodic properties.

REVISION



Electronic Configuration

a) ATOMS

1s 2s 2p 3s 3p 4s 3d 4p.....

s = two $\bar{e}$

p = six $\bar{e}$

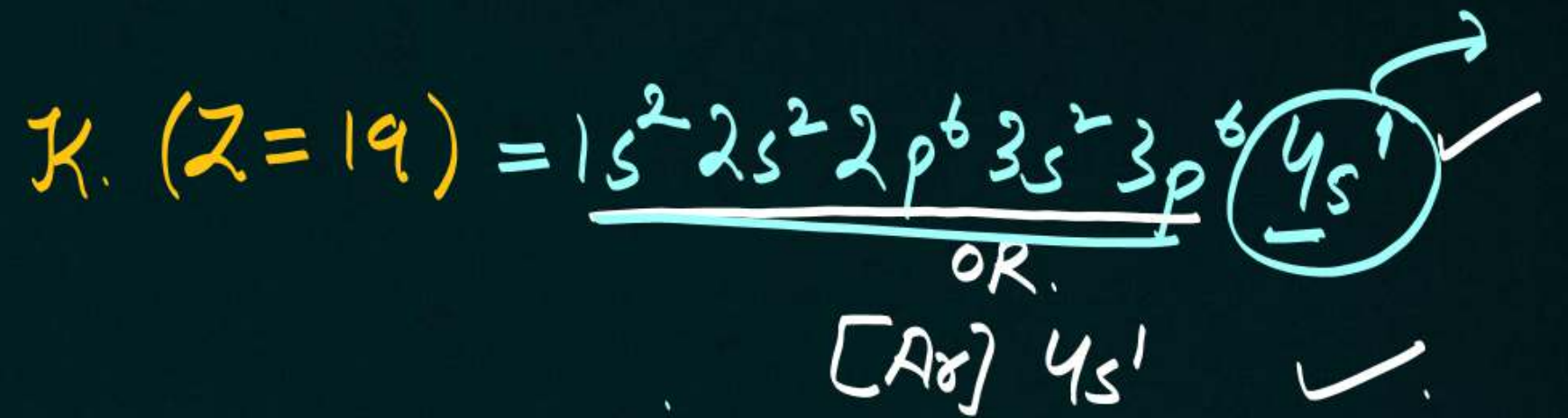
d = 10 $\bar{e}$

f = 14 $\bar{e}$

He = 2

Ne = 10

Ar = 18



Chapter.

1. SOME BASIC CONCEPTS OF CHEMISTRY.
 2. STRUCTURE OF ATOM.
 - 3) CLASSIFICATION OF ELEMENTS + PERIODIC PROPERTIES.
- } REVISION ✓

STABILITY OF HALF FILLED AND FULLY FILLED ORBITALS.

• Almost all the Elements follow Aufbau rule. except few elements like Cr & Cu. It is said that the gap between 4s & 3d subshell is very less i.e. why e jumps from 4s to 3d.

Electronic config. of -

a) Cr. (Z=24) $\rightarrow$ [Ar] 4s² 3d⁴ (Expected).
 $\rightarrow$ [Ar] 4s¹ 3d⁵ (Real electronic config.)

b) Cu. (Z=29) $\rightarrow$ [Ar] 4s² 3d⁹ (Expected)
 $\rightarrow$ [Ar] 4s¹ 3d¹⁰ (Real).

REASON $\rightarrow$

i) Symmetry $\rightarrow$ It is a universal rule that symmetry leads to stability. In case d⁵ & d¹⁰ there is symmetry in electrons.



ii) EXCHANGE ENERGY $\rightarrow$

Electrons with parallel spin within a subshell there occurs a exchange of position, & during this exchange, they release energy called. exchange energy. More the no. of exchanges, more is the released exchange energy, more is the stability.

Let us check the stability of d^4 & d^5



$$\text{Total exchange} = 3 + 2 + 1 = 6$$



$$\text{Total exchange} \rightarrow 4 + 3 + 2 + 1 = 10$$

Hence d^5 is more stable than d^4 .

NOTE.

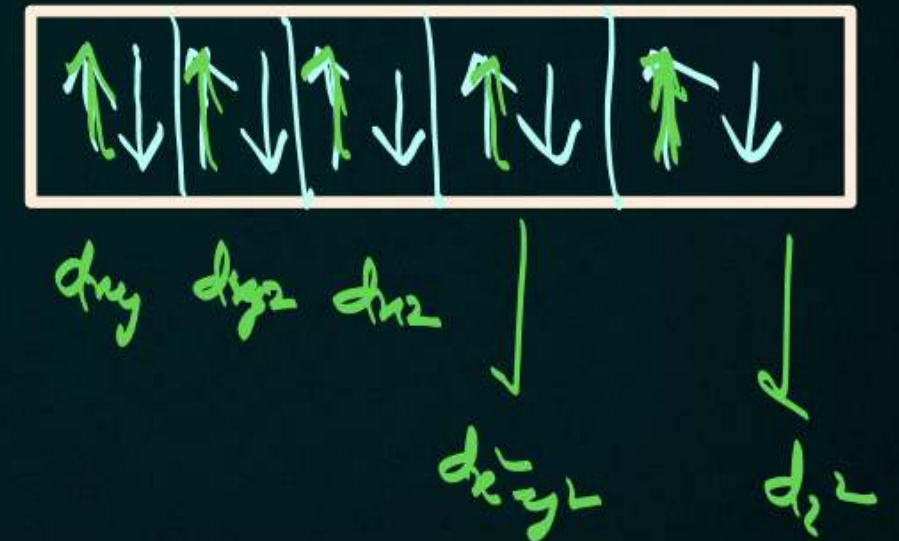
$$\text{No. of Exchange} = \frac{n(n-1)}{2}$$

n = no. of e^- with parallel spin.

Q. No. of exchange in p^3 or $\begin{array}{|c|c|c|} \hline \uparrow & \uparrow & \uparrow \\ \hline p_x & p_y & p_z \end{array}$

- a) 6 c) 4
- b) 3 d) 2

$$\begin{aligned} \text{No. of exchanges} &= \frac{n(n-1)}{2} \\ &= \frac{3(3-1)}{2} = \frac{3 \times 2}{2} = \frac{6}{2} = 3 \end{aligned}$$



Classification of element and periodicity in properties



Unit-III: Classification of element and periodicity in properties

6 Marks

- Significance of classification, brief history of the development of periodic table. Modern periodic law and the present form of the periodic table, periodic trends in properties of elements: atomic radii, ionic radii, inert gas radii, ionization enthalpy, electron gain enthalpy, electronegativity, valency.

#SIGNIFICANCE OF CLASSIFICATION OF ELEMENTS.

At the end of 17th century there were only 31 elements & at the end of 18th century there were 63 elements. Therefore it was easy to remember the name and properties of each elements. But today we have 118 elements, therefore now it is difficult to remember names and properties of each elements. Hence we need to classify the elements.

Classification of elements is defined as, grouping of elements with similar properties and separating the elements with dissimilar properties.

BRIEF HISTORY OF CLASSIFICATION OF ELEMENTS

1) DOBEREINER TRIADS

In 1817, JOHN DOBEREINER classified the then known elements in group of three elements called Triads. Elements in triad have similar properties and average of atomic weight of first and last element of a triad were equal to approximately atomic weight of middle elements.

Triads were.

- i) Li, Na, K.
- ii) Ca, Sr, Ba
- iii) Cl, Br, I.

Element	Atomic weight	
Lithium	7	Mean of 1st and 3rd $\frac{7 + 39}{2} = 23$
Sodium	23	
Potassium	39	
Calcium	40	Mean of 1st and 3rd $\frac{40 + 137}{2} = 88.5$
Strontium	88	
Barium	137	
Chlorine	35.5	Mean of 1st and 3rd $\frac{35.5 + 127}{2} = 81.25$
Bromine	80	
Iodine	127	

QUESTION

Why are half-filled and fully-filled orbitals extra stable?

- 1 High shielding effect
- 2 Low effective nuclear charge
- 3 Symmetry and high Exchange Energy
- 4 Small atomic radius

Lattu Sawal.

QUESTION

Aufbau principle does not give the correct arrangement of filling up the atomic orbitals in

1 Cu and Zn

2 Co and Zn ✓

3 Mn and Cr

4 Cu and Cr

Q. Calculate the no. of exchanges in d^3 .



a) 3

b) 4

c) 5

d) 6

$$\frac{n(n-1)}{2} = \frac{3(3-1)}{2} = \frac{3 \times 2}{2} = \frac{6}{2} = 3$$



$$\frac{n(n-1)}{2} = \frac{5(5-1)}{2} = \frac{5 \times 4}{2} = 10$$

$$\frac{s(s-1)}{2} = 10$$

$\begin{matrix} \textcircled{10} \uparrow \\ \textcircled{10} \downarrow \end{matrix} = 20$

Summary



- 1) Musaa
- 2) Mehran Ismail Bhat
- 3) Umar

REVISION KAR K SONA →

Thank
You